

**OBJECT-ORIENTED PROGRAMMING WITH APPLICATIONS TO NUMERICAL METHODS AND
LINEAR ALGEBRA, ES 4075 - 01
SPRING 2026, ASSIGNMENT – 2, # OF PROBLEMS: 3, TOTAL POINTS: 30
SUBMISSION DATE: 02/20/2026 BY 4:00 PM**

Submission:

- You need to submit a single zipped folder. Multiple file submissions will incur a 50% penalty.
- Code without appropriate comments will incur a 25% penalty.
- Submit problem 1 as a PDF.
- Submit the complete project folder for problems 2 and 3. The project that cannot be opened or cannot be compiled will incur a 50% penalty.

Prob. 1

Points: 10

Determine the roots of the non-linear equation shown below using the secant method with a **paper and pencil approach only (Should use only a calculator without any programs)**:

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0 \text{ with } x_1 = 0.05, \text{ and } x_0 = 0.02$$

Description of the Secant Method:

The Newton-Raphson method of solving the nonlinear equation $f(x) = 0$ is given by the recursive formula

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \quad (1)$$

From the above equation, a number of drawbacks of the Newton-Raphson method can be identified, and some of the drawbacks are listed below:

- The derivative of $f(x)$ has to be evaluated, which is a laborious process even with symbolic manipulators such as MATLAB and Mathematica.
- The derivative of $f(x)$ has to be evaluated when $f(x)$ is close to zero.

These two drawbacks are addressed by approximating the derivative by the difference quotient.

$$f'(x) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \quad (2)$$

Substituting eqn. (2) in eqn. (1), we have

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})} \text{ for } i \geq 1 \quad (3)$$

Eqn. 3 is called the secant method.

- The secant method requires two initial guesses, but unlike the bisection method, the two initial guesses need not bracket the root of the equation.
 - The secant method may or may not converge, but when it converges, it converges faster than the Bisection method and slower than the Newton-Raphson method since the derivative is approximated.
1. Show the steps of one iteration with a paper and pencil approach. The paper-and-pencil approach should first provide the algebraic equations, then the numerical values.

2. Provide the numerical values of four iterations in a tabular format as shown below:

Iteration	Approximate root	Error
1		
2		
3		
4		

Rules of Problems 2 and 3:

- You have to use the provided template projects.
- You should not define any new data variable unless stated in the comments.
- You are allowed to define loop variables.
- Do not alter or delete code provided in the templates.
- Write your code below the comments in the templates.
- The completed program's functionality, including output display, should be similar to the instructor's executables provided along with the template projects:

Prob. 2

Points: 10

Develop a program to implement the Secant method using the Problem2 template project.

- The template project includes comments that provide hints for code implementation.
- The instructor executable program is available in the zipped folder for determining the requirements of the program and validation.
- Demonstrate the program by determining the roots of the non-linear equation given below with a tolerance of 0.0001:

$$f(x) = x^3 - \sinh(x) + 4x^2 + 6x + 9 = 0 \text{ with } x_1 = 8.0, \text{ and } x_0 = 5.0$$

Prob. 3

Points: 10

Develop a program to compute the mean and standard deviation of a series of real numbers using the Problem3 template project.

- The template project includes comments that provide hints for code implementation.
- The instructor executable program is available in the zipped folder for determining the requirements of the program and validation.